

Self-checking testbench

A useful testbench does not stop at dumping waves for eyeball review

Starter and2 PASS

- Starter preset
- The flow row reads stimulus, DUT, got, compare, PASS
- The compare box shows y equals one matches expect one
- History lists one passing vector
- Try Run check again to refresh the TB sketch
- Set expect to zero on the same stimulus and Run check again

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Self-checking TB pattern

Pre-UVM literacy: **stimulus** → DUT → **expect** → compare → **pass/fail** (no eyeballing waves alone). Starter: and2 with a=1, b=1, expect y=1 — PASS.

Starter example: and2 with a=1, b=1, expect y=1, already checked — **PASS**.

Load starter example

Challenges 0 / 22 cleared

Quiz: goal: A self-checking TB's main job is...

☐ compare DUT outputs to an expected value and report pass/fail

☐ only dump VCD for manual review

☐ synthesize the DUT

☐ replace Accellera UVM

Show hint

Check

Next

Idle

1

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Core ideas

Stimulus

Drive DUT inputs from the TB.

Expect

Golden value from a reference model.

Compare

got !== exp → \$error (fail).

Tally

Count errors; exit with a clear result.

Self-check TB starter

Real RTL/TB practice

- In Track A, write the five-step self-check loop in order
- Sketch dollar-error versus dollar-display for one UART-style byte check, even on paper
- Optional: peek at a UART testbench in the legacy combined materials and name where expect
- This lab uses tiny DUTs; the discipline is the same for full serial VIP later

Pitfalls to watch

- Do not set expect by reading the DUT output you are testing
- Waves help debug but are not the pass criterion, automate the compare
- One passing vector is not coverage
- Wrong expect on purpose is good practice, you must see FAIL paths in sim logs
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, write the compare line for got not equal expect
- When you are ready, take the short quiz, then continue to TB clock and reset